

ENVS193DS-Final

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Repository Link

[GitHub Repository](#)

Set Up

Packages:

```
# reading in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(here)
library(scales)
library(ggeffects)
library(MuMIn)
library(forcats)
library(DHARMA)
library(ggeffects)
library(gtsummary)
```

Data:

```
# creating object from 'nests' from data
nests <- read.csv(here('data', 'nest_data_final.csv'))
```

Problem 1

a.

In part 1 of the results section, the co-worker used a linear regression. In part 2 of the results section, the co-worker used a Chi-Squared test.

b.

Statement 1: Based collected American Avocet micro-habitat use and distance to water edge, American Avocet micro-habitat use is significantly influenced by proximity to water edge.

Statement 2: Based on collected shorebird species presence across the habitats managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflap, shorebird species significantly differ by locale.

c.

One additional variable that could influence the probability of American Avocet microhabitat use is the water depth surrounding habitats, a continuous variable measured in meters, collected via bathymetry or hands-on-measurement depending on depth. This variable might influence American Avocet habitat use as water depth is a significant driver of bird use in managed wetlands – water depth may influence American Avocet use depending on feeding patterns (deeper areas are inaccessible to wading birds), present food type (different depths may lead to different flora and fauna presence), and predator presence (predators may reside in varying depths of water/predators may not be able to access certain water depths), playing a role in determining how American Avocets use these locales (Schacter et. al., 11).

Problem 2

a.

```
# create new object called 'nests_clean' from 'nests'
nests_clean <- nests |>
  # clean names
  clean_names() |>
  # filter ant species
  filter(
    ant_species %in% c("RRB", "AB", "BBR")) |>
  # full names
  mutate(
    species_name_full = recode_values(ant_species,
                                      "RRB" ~ "Crematogaster mimosae",
                                      "AB" ~ "Crematogaster sjostedti",
                                      "BBR" ~ "Crematogaster nigriceps")) |>
  # reordering species
  mutate(
```

```

# order of species
species_name_full = as_factor(species_name_full),
species_name_full = fct_relevel(species_name_full,
                                "Crematogaster mimosae",
                                "Crematogaster sjostedti",
                                "Crematogaster nigriceps")) |>

# arrange in order
arrange(species_name_full) |>
# columns of interest
select(
  height_cm,
  case_control,
  species_name_full)

# display structure
str(nests_clean)

```

```

'data.frame':  270 obs. of  3 variables:
 $ height_cm      : num  392 310 513 154 413 ...
 $ case_control   : int   1 0 0 0 1 0 0 0 0 0 ...
 $ species_name_full: Factor w/ 3 levels "Crematogaster mimosae",...: 1 1 1 1 1 1 1 1 1 1 ...

```

```

# display 10 random rows
nests_clean |> slice_sample(n = 10)

```

	height_cm	case_control	species_name_full
1	211.5	0	Crematogaster mimosae
2	131.0	0	Crematogaster mimosae
3	204.5	1	Crematogaster mimosae
4	249.0	0	Crematogaster mimosae
5	210.5	0	Crematogaster mimosae
6	144.0	0	Crematogaster mimosae
7	201.2	1	Crematogaster mimosae
8	90.5	0	Crematogaster mimosae
9	185.0	0	Crematogaster mimosae
10	159.0	0	Crematogaster mimosae

b.

In this data set, 1s and 0s indicate if a tree has a nest (1) or is “available” (0); in other words, they indicate if a tree contains a birds nest or not.

c.

Tree Height: Tree height is a continuous variable, measured in centimeters. According to paragraph three in the introduction subsection, as well as paragraph two in the results subsection of **Symbiotic acacia ants drive nesting behavior by birds in an African savanna**, as tree height increases, probability of nest occurrence probability will increase as well, as higher nests decrease ground predator nest predation rates, increasing overall bird lifetime fitness.

Acacia Ant Species: Acacia ant species is a categorical variable (C.nigriceps, C.mimosae, and C.sjostedti). According to paragraph three in the introduction subsection in **Symbiotic acacia ants drive nesting behavior by birds in an African savanna**, the presence of more aggressive defending symbionts (C.nigriceps and C.mimosae) will result in an increased probability of nest occurrence probability, aggressive defender presence may protect nests, reducing nest predation rates, and increase overall bird lifetime fitness.

d.

Model number	Ant Species	Tree Height	Predictor List
0 (null)			no predictors
1 (full model)	x	x	all predictors
2 (species)	x		species
3 (height)		x	tree height
4 (interact)	x	x	all predictors – reliance

e.

```
# model 0: null model
model_0 <- glm(
  # formula
  case_control ~ 1,
  # data frame
  data = nests_clean,
  family = "binomial")

# model 1: no interaction
model_1 <- glm(
  # formula
  case_control ~ height_cm + species_name_full,
  # data frame
```

```

    data = nests_clean,
    family = "binomial")

# model 2: ant species
model_2 <- glm(
  case_control ~ species_name_full,
  # data frame
  data = nests_clean,
  family = "binomial")

# model 3: tree height
model_3 <- glm(
  # formula
  case_control ~ height_cm,
  # data frame
  data = nests_clean,
  family = "binomial")

# model 4: interaction
model_4 <- glm(
  # formula
  case_control ~ height_cm * species_name_full,
  # data frame
  data = nests_clean,
  family = "binomial")

```

f.

```

# running AICc
AICc(
  model_0,
  model_1,
  model_2,
  model_3,
  model_4
) |>
  # display ordered results
  arrange(AICc)

```

The best model as determined by Akaike's Information Criterion (AIC) was the model using

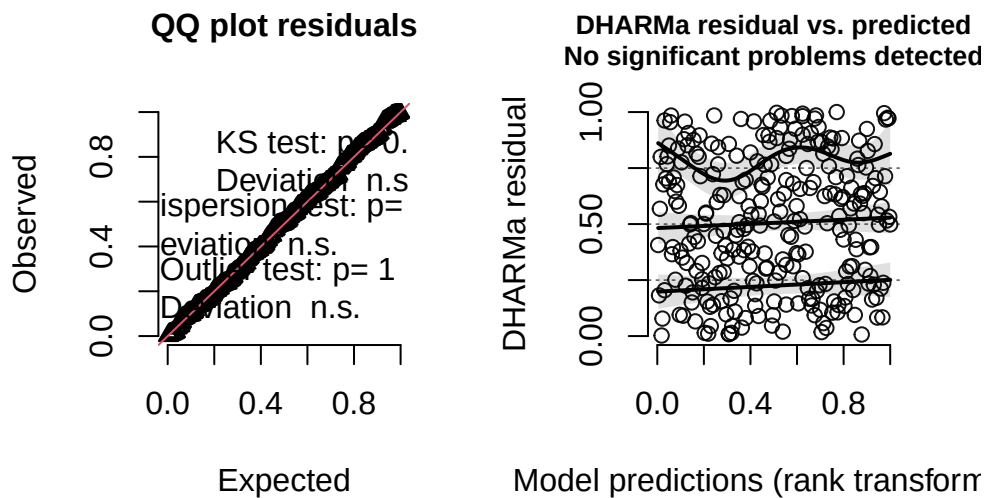
interacting predictor variables ant species and tree height for the response variable of bird nest presence (model 4).

g.

```
# running dharma diagnostic
diagnostic_checking <- simulateResiduals(fittedModel = model_4)

# display dharma diagnostic
plot(diagnostic_checking)
```

DHARMa residual



```
# display summary
summary(model_4)
```

Call:

```
glm(formula = case_control ~ height_cm * species_name_full, family = "binomial",
     data = nests_clean)
```

Coefficients:

	Estimate	Std. Error	
(Intercept)	-2.845076	0.434500	
height_cm	0.006492	0.001572	
species_name_fullCrematogaster sjostedti	-0.848334	2.554617	
species_name_fullCrematogaster nigriceps	-13.128072	5.405092	
height_cm:species_name_fullCrematogaster sjostedti	-0.002597	0.008386	
height_cm:species_name_fullCrematogaster nigriceps	0.081944	0.030921	
	z value	Pr(> z)	
(Intercept)	-6.548	5.83e-11	***
height_cm	4.130	3.63e-05	***
species_name_fullCrematogaster sjostedti	-0.332	0.73983	
species_name_fullCrematogaster nigriceps	-2.429	0.01515	*
height_cm:species_name_fullCrematogaster sjostedti	-0.310	0.75679	
height_cm:species_name_fullCrematogaster nigriceps	2.650	0.00805	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 286.04 on 269 degrees of freedom
Residual deviance: 225.80 on 264 degrees of freedom
AIC: 237.8

Number of Fisher Scoring iterations: 7

h.

```
# generating all predictions
predictions <- ggpredict(
  model_4,
  terms = c("height_cm [all]",
            "species_name_full")) |>
# renaming columns
rename(height_cm = x,
        species_name_full = group)
```

```
# base layer: ggplot
# use nests_clean data
ggplot(data = nests_clean,
  # x-axis
  aes(x = height_cm,
```



```

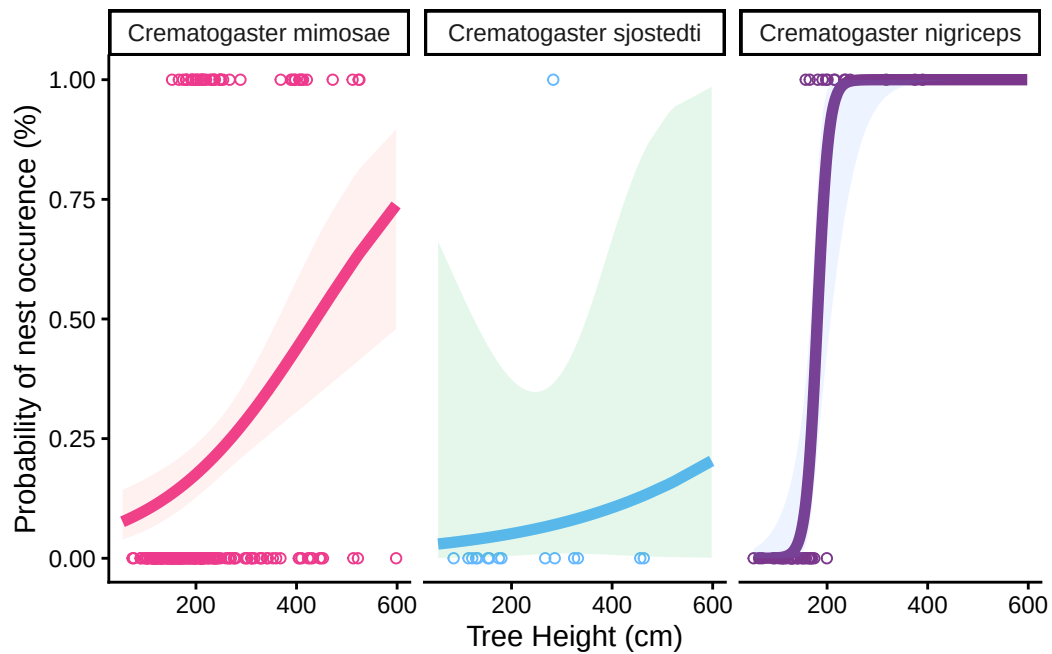
    # y-axis
    y = case_control)) +
# first layer: points
# color by species
geom_point(aes(color = species_name_full),
           # change point shape
           shape = 21) +
# second layer: line
# use predictions data
geom_line(data = predictions,
          # x-axis
          aes(x = height_cm,
              # y-axis
              y = predicted,
              # color by species
              color = species_name_full),
          # set linewidth
          linewidth = 2) +
# third layer: ribbon
# use predictions data
geom_ribbon(data = predictions,
           # x-axis
           aes(x = height_cm,
               # y-axis
               y = predicted,
               # y-min
               ymin = conf.low,
               # y-max
               ymax = conf.high,
               # fill by species
               fill = species_name_full),
           # change opacity
           alpha = 0.1) +
# panels by species
facet_wrap(vars(species_name_full)) +
labs(
  # label x-axis
  x = "Tree Height (cm)",
  # label y-axis
  y = "Probability of nest occurrence (%)" +
# color by species
scale_color_manual(

```

```

values = c("Crematogaster mimosae" = "violetred2",
           "Crematogaster sjostedti" = "steelblue1",
           "Crematogaster nigriceps" = "mediumorchid4")) +
# use classic theme
theme_classic() +
# remove legend
theme(legend.position = "none")

```



i.

j.

```

# generating prediction
ggpredict(model_4,
          # setting terms
          terms = c("height_cm [600]",
                    "species_name_full"))

```

```

# Predicted probabilities of case_control

```

species_name_full: *Crematogaster mimosae*

height_cm	Predicted	95% CI
600	0.74	0.48, 0.90

species_name_full: *Crematogaster sjostedti*

height_cm	Predicted	95% CI
600	0.20	0.00, 0.99

species_name_full: *Crematogaster nigriceps*

height_cm	Predicted	95% CI
600	1.00	1.00, 1.00

k.

Problem 3

a.

My visualizations use different metrics and different model types across Homework 2/3 and my affective visualization; in Homework 2 and 3 I used box plots to visualize my data, while I used a trendline to visualize my data in my affective visualization. Across my visualizations, I did not see many similarities – my affective visualization reflected my predicted outcomes with different variables, while my visualizations in homework 2/3 did not show the trends that I depicted in my affective visualization; similarly, I did not see many patterns across my affective and effective visualizations, likely due to different metrics of creation and data alignment. I did not receive any feedback on my Homework 2 and 3 visualizations, however, in workshop, I got a lot of visual suggestions – for instance, implementing a clearer trend line in my drawing – which was reflected in my final affective visualization (I added a red trend line, and used colors to create clearer visual distinctions between my data).

b.

Using the response variable of productivity % and the predictor variable of activity time (as expressed in my affective visualization as an aggregation of outside activity and class time), one analysis that we talked about this quarter that I could apply to my research question

is a correlation analysis. Both of my variables are numerical continuous variables, with one variable (activity time) predicted to have an influence on the other variable (productivity %). Subsequently, I believe that a correlation analysis is appropriate in trying to answer the question: how do my activities impact my productivity?

c.

d.

e.

f.

g.

Completed in section